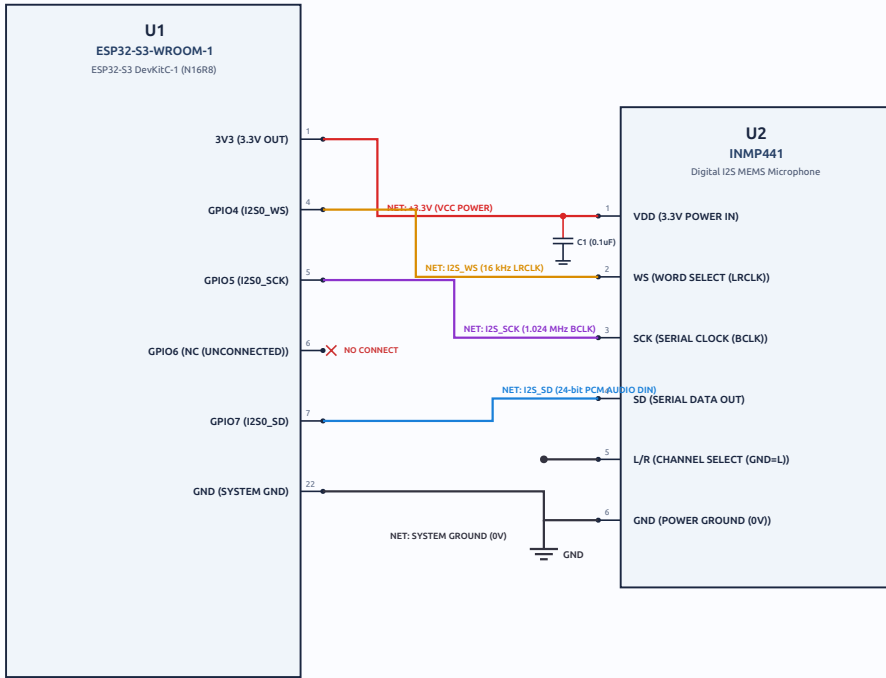


SECTION 1: ELECTRICAL LOGIC SCHEMATIC (EDA CIRCUIT VIEW)



SECTION 2: PHYSICAL PCB FOOTPRINT & HARDWARE BENCH WIRING



SECTION 3: HARDWARE INTERFACE SPECIFICATION & PINOUT MATRIX

INMP441 PIN	ESP32-S3 PIN	WIRE COLOR	SIGNAL NAME	ELECTRICAL & PROTOCOL SPECIFICATION	STATUS / VERIFICATION
VDD	3V3 (Pin 1)	RED	+3.3V DC Power	Strict 3.3V supply (operating 1.62V - 3.63V). NEVER connect to 5V!	ACTIVE (Firmware Verified)
GND	GND (Pin 22)	BLACK	System Ground (0V)	Common return rail. Connect directly to breadboard GND bus.	ACTIVE (0V Reference)
L/R	GND Rail	GRAY / WHITE	Channel Select	Tied LOW to Ground -> Microphone transmits on LEFT channel.	ACTIVE (Pulled LOW)
WS	GPIO 4	YELLOW	Word Select (WS)	16 kHz Frame Synchronization. Left channel active when WS = 0.	ACTIVE (I2S_NUM_0)
SCK	GPIO 5	PURPLE	Serial Clock (SCK)	Continuous Bit Clock (BCLK) = 1.024 MHz (16 kHz x 32 bits x 2 ch).	ACTIVE (Master Mode)
SD	GPIO 7	BLUE	Serial Data Out	24-bit PCM digital data shifted on falling edge of SCK. Internal Pulldown.	ACTIVE (DMA Ring Buf)

ORGANIZATION / EVENT:
SMART INDIA HACKATHON 2026

DRAWING TITLE:
ESP32-S3 TO INMP441 I2S SCHEMATIC
EDGE-TO-CLOUD VOICE ASSISTANT PIPELINE

DOC NUMBER:

SIH-HW-001

REVISION:

REV 2.2 (PROD)

SHEET SIZE:

A4 LANDSCAPE

DATE: 2026-09-08

SCALE: 1:1 PCB / NTS EDA

DRAWN BY: Hardware Team

STATUS: VERIFIED READY

SHEET: 1 OF 1

PRINT RESOLUTION: 300 DPI